

Assignment 2: Exploratory Data Analysis

$$1. m(a+bX) = \frac{1}{N} \sum_{i=1}^N (a+bX_i) = \frac{1}{N} \sum_{i=1}^N a + \frac{1}{N} \sum_{i=1}^N (bX_i) = \frac{1}{N} \cdot (a \cdot N) + b \cdot \underbrace{\left(\frac{1}{N} \sum_{i=1}^N X_i \right)}_{m(X)} = \boxed{a + b \cdot m(X)}$$

$$\begin{aligned} 2. \text{COV}(X, a+bY) &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (a+bY_i - \underbrace{m(a+bY)}_{\text{from 1, this is } a+b \cdot m(Y)}) \\ &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (a+bY_i - a - b \cdot m(Y)) \\ &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (bY_i - b \cdot m(Y)) \\ &= b \cdot \underbrace{\frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (Y_i - m(Y))}_{\text{COV}(X, Y)} \\ &= \boxed{b \cdot \text{COV}(X, Y)} \end{aligned}$$

$$\begin{aligned} 3. \text{COV}(a+bX, a+bX) &= \frac{1}{N} \sum_{i=1}^N \underbrace{(a+bX_i - m(a+bX))}_{a+b \cdot m(X)} \underbrace{(a+bX_i - m(a+bX))}_{a+b \cdot m(X)} \\ &= \frac{1}{N} \sum_{i=1}^N (a+bX_i - a - b \cdot m(X)) (a+bX_i - a - b \cdot m(X)) \\ &= \frac{1}{N} \sum_{i=1}^N (bX_i - b \cdot m(X)) (bX_i - b \cdot m(X)) \\ &= b \cdot b \cdot \underbrace{\frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (X_i - m(X))}_{\text{COV}(X, X)} \\ &= \boxed{b^2 \cdot \text{COV}(X, X)} \end{aligned}$$

$$\text{COV}(X, X) = \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (X_i - m(X)) = \frac{1}{N} \sum_{i=1}^N \underbrace{(X_i - m(X))^2}_{s^2} = \boxed{s^2}$$

4. No, it is not always true. It only holds true if the function $g(\cdot)$ is linear.